BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY
Keyspan Gas East Corporation d/b/a National Grid

Cases 23-G-0225 & 23-G-0226

September 1, 2023

Prepared Testimony of:

Staff Gas Reliability & Supply Panel

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Q. Members of the Gas Reliability & Supply Panel, or SGRSP, please state your names, employer and business address.

A. Our names are Davide Maioriello and Jeffrey Kline. We are employed by the New York State Department of Public Service, referred to as the Department. Our business address is Three Empire State Plaza, Albany, NY 12223.

Q. Mr. Maioriello, please state your position at the Department.

A. I am a Utility Supervisor within the Gas System Planning and Reliability Section of the Office of Energy System Planning & Performance.

Q. Please summarize your educational and professional experience.

A. I attended Hudson Valley Community College and graduated with an Associate degree in Applied Science in construction technology. I continued my education at the SUNY Institute of Technology and, in May 1999, graduated with a Bachelor of Science degree in civil engineering technology. From February 2000 to February 2003, I was employed by SPEC Consulting, LLC as a Project Engineer. My work involved project management

and coordination, engineering, as well as various computer-aided design projects. From February 2003 to December 2005, I worked for the City of Albany Engineering Division where I was responsible for several tasks, including inspections of major roadway projects and other municipal construction projects. I also engaged in permit processing and utility work inspections. In December 2005, I joined the Department and since that time I have performed various engineering analyses of projects and associated capital budgets, operations and maintenance, referred to as O&M, expenses and programs, distribution system reliability needs, capacity and demand requirements, and review of various petitions and tariff filings of gas and electric utility companies in New York State.

Q. Please describe your duties in the Office of Energy System Planning and Performance.

A. My responsibilities within the office focus on the analysis of natural gas utility policy issues, including distribution system reliability and planning, capacity asset management, gas purchasing practices, gas system

reliability, and issues related to the restructuring of the natural gas industry and use of natural gas in New York State.

Q. Have you previously provided testimony in a proceeding before the New York State Public Service Commission, or Commission?

A. Yes. I have provided testimony with respect to a number of issues, including non-pipeline alternatives, referred to as NPAs, capital expenditures, materials and supplies, depreciation rates, sales forecasting, revenue decoupling mechanisms, revenue allocation, rate design, lost and unaccounted for gas, capital expenditure reconciliation mechanisms, retail access issues, reliability forecasting and supply planning, peaking supply and capacity issues, design day and winter forecasting, reliability improvements, gas network enhancement and reporting requirements. I have provided testimony in the following rate proceedings: Cases 22-G-0065 and 19-G-0066, regarding Consolidated Edison Company of New York, Inc., or Con Edison; Cases 05-G-1494 and 14-G-0494, regarding Orange and Rockland

Utilities, Inc.; Cases 05-G-1635, 08-G-1392 and 15-G-0382, regarding St. Lawrence Gas Company, Inc.; Cases 06-G-1185, 16-G-0059 and 19-G-0309, regarding The Brooklyn Union Gas Company d/b/a National Grid NY, or KEDNY; Cases 19-G-0309, 06-G-1186, 16-G-0058 and 19-G-0310, regarding Keyspan Gas East Corporation d/b/a National Grid, or KEDLI; Cases 07-G-0141 and 16-G-0257, regarding National Fuel Gas; Cases 08-G-0609, 12-E-0201, 12-G-0202, 17-G-0239 and 20-G-0381, regarding Niagara Mohawk Power Corporation; Case 09-G-0716 and 19-G-0379, New York State Electric & Gas Corporation, referred to as NYSEG; Cases 19-G-0381 and 09-G-0718, regarding Rochester Gas and Electric Corporation, referred to as RG&E; and Case 14-G-0319, regarding Central Hudson Gas & Electric Corporation.

Q. Mr. Kline, please state your position at the Department.

A. I am employed by the Department as a Utility Supervisor (Safety) in the Pipeline Safety Section of the Office of Energy System Planning & Performance.

Q. Mr. Kline, have you provided other testimony in

these proceedings?

A. Yes. I provided testimony as part of Staff's Pipeline Safety Panel and my credentials are listed in that testimony.

Q. How will you refer to the Brooklyn Union Gas Company d/b/a National Grid and KeySpan Gas East Corporation d/b/a National Grid in your testimony?

A. We will refer to the Brooklyn Union Gas Company d/b/a National Grid as KEDNY, and KeySpan Gas East Corporation as KEDLI. We will refer to KEDNY and KEDLI collectively as the Companies.

Q. Panel, what is the purpose of your testimony in these proceedings?

A. The Panel's testimony will address the Gas Planning Proceeding and Staff's concerns related to the following proposals made by KEDNY and KEDLI: 1) Gas Planning Proceeding; 2) Renewable Natural Gas, or RNG, supply; 3) Liquified Natural Gas, or LNG, Infrastructure and Supply; 4) the Interruptible Service Classes; 5) Utility Thermal Energy Network Pilot; and 6) Hydrogen Pilot, or HyGrid Project.

Q. In this testimony, will the Panel refer to, or

otherwise rely upon, any information produced during the discovery phase of this proceeding?

A. Yes. The Panel will refer to, and rely upon, several responses to Information Requests referred to as IRs, all of which are included in Exhibit___(SGRSP-1). We will refer to them by the designation given to them by the Department, for example, DPS-100.

Q. Are you sponsoring any other exhibits in support of your testimony?

A. No, the panel is not sponsoring any other exhibits.

Q. What is the Rate Year in these proceedings?

A. The Rate Year is the 12-month period ending March 31, 2025.

Gas Planning Proceeding

Q. Have any new Commission directives been issued with regard to the future of gas system planning since the Companies' last rate filing?

A. Yes. The Commission issued an Order Adopting Gas System Planning Process, referred to as the Planning Order, on May 12, 2022, in Case 20-G-0131, Proceeding on Motion of the Commission in Regard to Gas Planning Procedures, or the Gas

Planning Proceeding. In the Planning Order, the Commission established a new natural gas system planning process for the major gas utilities in the State, also referred to as Local Distribution Companies, or LDCs.

Q. Please summarize the new planning requirements.

A. The modernized gas system planning process requires that each LDC develop and submit substantive long-term system plans, referred to as long-term plans. The process requires each LDC to file an initial long-term plan on a staggered schedule and thereafter on a three-year cycle. The Planning Order established the following requirements for the long-term plans: 1) the long-term plans must use a 20-year horizon; 2) stakeholder education and participation requirements as part of the collaborative review of each LDC's long-term plan; 3) independent third-party consultant review of LDC filings; and, 4) substantive requirements for developing long-term plans which includes, among other things, the implementation of an NPA Screening process and framework. Further, the Planning Order requires

each LDC to file an annual report in each of the years between its full long-term plans. The annual reports must include actual system data and progress on the LDC's recent long-term plan and additional information to help stakeholders, Staff, and the Commission continue to develop and maintain their awareness and understanding of the LDC's long-term plan.

Q. Does the Commission's Order have any additional requirements?

A. Yes, in recognition of participant comments and concerns, the Planning Order identified next steps for issues that remain to be addressed in the Gas Planning Proceeding. Among them, the Commission recognized that continued extension of natural gas mains may be contrary to efforts to achieve the Climate Leadership and Community Protection Act, or CLCPA, carbon and greenhouse gas, or GHG, emission reduction targets and directed each LDC to file a report on the costs related to the extension of gas service, commonly known as the 100-foot rule contained in 16 New York Codes, Rules, and Regulations Part 230, and related data.

Q. How will the report inform the Commission's efforts?

A. The report will be used as a guide for the development of proposed revisions to 16 New York Codes, Rules, and Regulations Part 230 which sets forth the rights and responsibilities of gas utilities and applicants for gas service regarding the extension of facilities.

Q. Did the Commission, by its Order, direct LDC's to perform any additional work or file any other studies or reports?

A. Yes. The Commission directed the LDCs to file depreciation studies to examine the structure of accelerated depreciation and impacts to ratepayers. Also, the Commission directed Staff to convene an Avoided Cost of Gas Working Group, open to all interested parties, to address identified differences and issues between electric and gas benefit cost analyses. This working group is expected to produce a report recommending modifications for future Commission consideration. Finally, the Commission highlighted other topics that may be included in future phases of the proceeding once more

guidance and recommendations are issued from the New York State Climate Action Council, or CAC, by 2023.

Q. When are KEDNY and KEDLI expected to file their initial LTP?

A. They are required to file their initial long-term plan by May 31, 2024.

Q. Does the Panel recommend the Companies, or the Commission pursue any additional actions in these rate proceedings related to the Gas Planning Proceeding or their LTP?

A. We do not recommend any further action as part of these proceedings, however, the Commission in its final rate order in these proceedings should reflect any changes developed out of the Gas Planning Proceeding if known by that time.

Renewable Natural Gas Supply

Q. Please define RNG as a renewable energy resource.

A. As stated in the U.S. Energy Information Administration's Glossary, renewable energy resources are those that are naturally replenishing but are flow-limited. Renewable energy resources include biomass, hydro,

geothermal, solar, wind, ocean thermal, wave action, and tidal action. RNG supplies are those produced from renewable biomass, such as the gas product from landfills and anaerobic digesters and are therefore a renewable energy resource. Since RNG is created as a byproduct of decomposition, RNG is virtually inexhaustible in duration but limited in the amount of energy that is available at any point in time.

Q. Why is it useful for the Companies to implement interconnection facilities for RNG sources within their service territories?

A. RNG sources can help offset the need for additional upstream pipeline capacity and provide additional localized supply within the distribution system. These sources also have the potential to reduce methane emissions, which would help to achieve the CLCPA's goals of a 40 percent reduction in GHG emissions from 1990 levels by 2030 and 85 percent by 2040. Additionally, the CAC released its Final Scoping Plan on December 19, 2022. On page 350, the CAC highlighted the need to further study the use of RNG and recognized that the State must explore

whether full electrification of heating load in the near term is the most cost effective and technically feasible solution for all customers. Further, the CAC noted that alternative fuels may have a role in the transition of the natural gas industry.

Q. Further, is it useful for the Companies to implement the interconnection facilities for RNG sources to address capacity constraints?

A. Yes. The CAC's Final Scoping Plan, on page 361, also recognized the role that renewable fuels may play "to meet customer needs for space heating or process use where electrification is not yet feasible or to decarbonize the gas system as it transitions." The CAC's recognition of renewable fuels as playing an important part in the transition to electrification pairs with the Companies' need to address its own capacity constraints.

Q. Does the Panel have any recommendations in addressing areas with capacity constraints?

A. Yes. Considering the emission reductions required by the CLCPA and the uncertainty associated with siting new pipeline projects,

the Panel recommends that the Commission require the Companies to continue their solicitation processes and pursue opportunities for additional NPAs in areas of their service territory with capacity constraints, such as with RNG projects.

Q. Are the Companies pursuing any potential RNG projects for interconnection opportunities for gas supplies to their distribution systems?

A. Yes. As described in further detail on page 78 of KEDNY Gas Infrastructure and Operations Panel's, or GIOP's direct testimony, KEDNY proposes two RNG Interconnection Projects with an estimated cost of $5.65 million for RNG Interconnection Project 1, and $8.4 million for RNG Interconnection Project 2. In addition, on page 74 of KEDLI's GIOP's direct testimony, KEDLI proposes two RNG Interconnection projects with an estimated cost of $5.5 million for KEDLI's Interconnection Project 1, and $5.7 million for KEDLI's RNG Interconnection Project 2.

Q. What are the Companies' proposed capital expenditures during the Rate Year?

A. The Companies are pursuing a total of four interconnections and forecast expenditures of $2.16 million and $2.66 million in the Rate Year, for KEDNY and KEDLI respectively. Additionally, the Companies have requested 1.5 FTEs for each company.

Q. Does the Panel have any concerns regarding the Companies' proposed RNG Interconnection Projects?

A. Yes, our concern is that these developer-led RNG projects are still in the very early stages of development as highlighted in the Companies' response to DPS-354. There is a high degree of likelihood that implementation and the in-service date of the interconnection facilities will be delayed.

Q. Is this risk consistent with the risks observed regarding the Companies' other or previous RNG projects?

A. Yes, this is similar to what has occurred since the Companies last rate proceedings, in Cases 19-G-0309 and 19-G-0310, where the Companies have not implemented any of the potential RNG projects tied to developer's proposals to date.

Q. Does the Panel have recommendations to address the risks associated with the implementation of these RNG projects?

A. Yes, a different cost recovery mechanism is needed to provide flexibility because of the inherent uncertainty of developer-led projects where the Companies are reliant on developers to actually coordinate, site, construct, and implement such RNG facilities and then coordinate with the Companies to construct and implement the gas system interconnection to safely interconnect and accept the RNG gas supply into their systems.

Q. What does the Panel recommend?

A. The Panel recommends removing these RNG projects from the Companies' proposed capital budgets, and projected plant-in-service balances. However, we recommend that the Commission allow the Companies to recover the revenue requirement impact of these projects through a surcharge mechanism if they are built and placed into service during the Rate Year. The surcharge mechanism would allow the Companies' to recover the revenue requirement impacts, i.e., return on

and depreciation expense, of the interconnection facilities. We also recommend this surcharge mechanism be limited to the revenue requirement impacts associated with the current estimated costs of these projects. The current estimated costs for these projects are shown in the Companies Exhibit___(GIOP-1).

Q. Does the Panel recommend any requirements prior to recovery through the surcharge?

A. Yes, we recommend that the Commission include requirements consistent with those it instituted in the recent Consolidated Edison rate proceeding, Cases 22-E-0064 and 22-G-0065, in its Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans with Additional Requirements on page 128. In the Commission's Order it adopted additional requirements, among others, that required Con Edison file a report 90 days prior to the construction of any interconnection project to receive RNG, which would contain the following information: a cost estimate of the interconnection project; a summary of benefits to the reliability of the natural gas system in

the vicinity of the interconnection project and in the service territory in general; a detailed description of the source materials being used at the interconnected facility to produce the RNG; and a detailed accounting of the upstream GHG emissions avoided by the RNG that the Companies will procure as a result of the interconnection. Thus, we recommend the Commission adopt consistent requirements for the Companies' RNG projects.

Q. Does the Panel have any other recommendations concerning these RNG Projects?

A. Yes, consistent with the Commission's requirements of Con Edison, the Companies' recovery of the costs associated with the purchase of RNG, as with all other purchases of natural gas supply should and will be accomplished through the Gas Adjustment Statement mechanism and such costs would be reviewed by DPS Staff as part of its annual review of gas costs. As part of its review, DPS Staff reports any concerns identified in its review to the Commission. Additionally, the Panel has concerns with the cost of RNG supplies

and recommends that RNG pricing should not be at a premium to the traditional supply portfolio unless it is implemented as an NPA solution that is providing quantified system benefits, and offsetting traditional capital projects, supporting such a premium cost to customers.

Q. Should the Companies purchase the environmental attributes associated with the RNG?

A. The Companies should not purchase the environmental attributes associated with RNG. Doing so would expose customers to unnecessary price volatility and potentially high costs for the commodity.

Q. Does the Panel have any recommendations regarding the requested FTEs?

A. Due to the very limited number of potential RNG projects and the implementation risks discussed earlier we do not support the request for additional staffing at this time and we therefore removed the cost the three FTEs from the Companies' revenue requirements.

LNG Facilities and Capacity Concerns

Q. Are the Companies proposing any LNG projects?

A. Yes. As described in more detail on page 68 and

page 64 of the of KEDNY and KEDLI's Gas Infrastructure and Operations Panels' direct testimony, respectively, the Companies discuss specific LNG capital improvement projects and their respective "blanket" LNG programs.

Q. Does the Panel have concerns with the Companies' LNG programs and proposals?

A. Yes, the LNG facilities for KEDNY and KEDLI are a major source of on-system natural gas supply capable of reliably meeting peak or design day demand. Our concern is that if the LNG facilities for either service area were inoperable during the historically coldest part of the year or on a peak cold weather day it would create the potential for firm customer curtailments. Each LNG facility provides natural gas supply and pressure support during cold weather events when firm customers need the peaking gas and high customer demands may reduce system pressures, impacting deliverability, and upstream pipeline capacity cannot provide enough supply to meet firm customers' demand.

Q. How much gas supply capability can the LNG facilities provide?

A. As indicated in the KEDNY's GIOP's direct testimony, the Greenpoint LNG Plant, can provide up to 290 million cubic feet, or about 300 million dekatherms, per day of gas supply directly into the KEDNY service area. As indicated in KEDLI's GIOP's direct testimony, the Holtsville LNG Plant, can provide up to 100 million cubic feet, or about 103 million dekatherms, per day of gas supply directly into the KEDLI service area.

Q. Does the Panel have any recommendations regarding the LNG facilities?

A. The Panel supports the Companies' proposed capital programs associated with the LNG facilities due to the very significant amount of peak day supply that each LNG facility provides for their respective service areas and the fact that other supply sources cannot be used as a substitute for the LNG facilities due to ongoing capacity constraints in each Company's service area. The LNG facilities capital programs will provide necessary improvements to the facilities to maintain reliability, the safety of each facility, and ensure operability when called

upon during the winter heating season.

Interruptible Service Classes

Q. What concerns do you have with the Companies' interruptible service classes?

A. The Panel is concerned with the future participation level of customers remaining on this service class offering. In its response to DPS-353, included in Exhibit___(SGRSP-1), the Companies state that these non-firm service classes have lost a total of 235 non-firm customers over the past three calendar years. Customer enrollment in this type of service provides a very important demand reduction benefit to each of the Companies' service areas. According to the response to DPS-988, included in Exhibit___(SGRSP-1), these non-firm service classes provide a demand reduction to the design peak day of approximately seven percent for KEDNY and approximately one percent for KEDLI.

Q. Did the Companies make any proposals regarding interruptible service?

A. Yes, as described in more detail on page 131 of the Customer Panel's direct testimony, the Companies propose to increase the discount off

the otherwise applicable firm rate by an additional five percent. Additionally, the Companies request an FTE to focus program outreach efforts to participating non-firm customers to reduce non-firm customer migration to firm service.

Q. Does the Panel have any recommendations regarding the Companies' interruptible service class proposals?

A. Yes, we support the Companies' proposals because these interruptible service customers provide a major peak reduction benefit on behalf of firm customers and help maintain necessary system pressures for reliable supply during peak cold weather events. If the Companies had to serve all non-firm customers on a firm basis, it would require adding infrastructure and contracting for more upstream capacity and supply assets. Therefore, we recommend the Commission adopt the Companies' proposal to increase the Interruptible Service Class discount and to hire an additional FTE.

Utility Thermal Energy Networks

Q. What is a Utility Thermal Energy Network, or

UTEN, pilot project?

A. A UTEN pilot project is a project designed to test the feasibility and business model of utility owned thermal networks. In its Order on Developing Thermal Energy Networks Pursuant to the Utility Thermal Energy Network and Jobs Act, or UTEN Order, issued on September 15, 2022, in Case 22-M-0429, the Commission initiated that proceeding to consider the Companies', amongst other utilities, UTEN pilot projects and related matters.

Q. What are the Companies proposing regarding UTEN pilot projects in these rate cases?

A. As discussed in more detail on page 85 of both the KEDNY and KEDLI GIOP's direct testimonies, the Companies propose to hire five FTEs for each Company, associated with its UTEN pilot projects as proposed in Case 22-M-0429.

Q. What concerns do you have with the Companies request for new personnel staffing in this rate proceeding?

A. Although we understand that there may be a need for the staffing positions, we recommend removal of all of the proposed FTEs associated with the

UTEN pilot projects in these proceedings. We recommend that the Companies address all costs associated with the UTEN pilot projects in Case 22-M-0429.

Q. Why is the Panel recommending removal of these requested FTEs?

A. In its UTEN Order, the Commission stated that the UTEN filings "for each proposed pilot project imposing incremental costs on the utility is to include a proposed cost recovery mechanism with an explanation regarding why the proposed mechanism is reasonable under the circumstances." Therefore, the Companies should address the costs of the UTEN pilot projects, including for FTEs and propose appropriate cost tracking and recovery mechanisms in Case 22-M-0429.

Hydrogen Pilot

Q. Please describe KEDLI's hydrogen pilot project.

A. As described in further detail on page 81 of the direct testimony of KEDLI's GIOP, KEDLI is proposing a project to expand an existing green hydrogen facility located in the Town of Hempstead. The project, also referred to as the

HyGrid Project, consists of an expansion of the compressed hydrogen storage system currently connected to an existing hydrogen vehicle fueling station. The pilot would expand the storage system, add an electrolyzer for hydrogen production, and construct an interconnection facility for KEDLI to blend hydrogen into the local gas distribution system in the Hempstead area. KEDLI is requesting a total of $13.1 million in capital costs and requesting a total of 1.5 FTEs for KEDLI and 0.5 FTEs for KEDNY as highlighted in KEDNY Exhibit___(GIOP-8).

Q. Is the Panel familiar with whether and at what rate hydrogen blended gas would leak from the Companies gas system?

A. Due to the nascent nature of hydrogen blending programs in New York State the Panel is not familiar with the permeability of existing in-service pipeline materials by hydrogen and to what extent it poses a potential ignition hazard. Further detailed study is required to establish permeability rates and if the hydrogen losses over distances could accumulate into concentrations for ignition to occur.

Q. What is the ignition range for hydrogen gas?

A. Relative to natural gas, hydrogen has a wider ignition range. For example, natural gas typically has an ignition range of between five and 15 percent concentration of gas-in-air, whereas hydrogen's ignition range is between four and 74 percent gas-in-air.

Q. Why is the ignition concertation range of hydrogen in air important?

A. Like natural gas, hydrogen gas is odorless and colorless. Under the Commission's pipeline safety regulations, natural gas in distribution systems must be odorized so as to render it readily detectable by the public at a concentration of one tenth of the lower ignition level. As hydrogen has a much wider ignition range, leaks on a gas system carrying a blend of natural gas and hydrogen represent a potentially greater risk to the public than leaks on a gas system solely carrying natural gas. Depending on its composition, a hydrogen-natural gas blend will have a lower explosive limit somewhere in between that for hydrogen and natural gas. In addition, the wide gap between upper limits

further compounds the issue of what is the explosive range for the blend. Knowing the explosive range is key to knowing when a leak is dangerous and helps determine actions when responding to the leak. Therefore, the blended gas should be odorized to be readily detachable at a concentration of one tenth of its lower explosive level.

Q. Are there concerns about the use of odorants?

A. It is critical that the odorant used does not interact with the hydrogen blended gas so as to render the odorant ineffective, become toxic, or cause harm to the integrity of the system.

Q. What is known about the interaction of odorants and hydrogen-natural gas blends?

A. There are limited studies on the impact of hydrogen blended into natural gas as it pertains to odorants. Therefore, we do not support blending on segments of the system with existing known leaks or on pipeline materials known to be leak prone.

Q. Does the Panel have any other concerns regarding hydrogen blended gas?

A. Yes, we are concerned with pipeline integrity

and leak detection.

Q. What concern does the Panel have related to pipeline integrity?

A. Excess hydrogen in pipelines is known to cause embrittlement of pipeline materials. Embrittlement leads to cracking, which in turn can lead to a leak or rupture of the pipeline. Additionally, the effect of hydrogen concentrations on gaskets, o-rings, and diaphragms used in the existing gas system is not fully known or understood. Introduction of hydrogen into an existing gas system could result in these components failing, which could cause a leak. Without understanding repair methods' susceptibility to be leak prone with respect to hydrogen, the Panel does not recommend that a pilot project be on a system with repaired leaks.

Q. What concern does the Panel have related to leak detection?

A. We are also concerned about being able to detect and quantify leaks using current leak detection devices which quantify leaks based on gas-in-air or as percent of lower explosive limit. Leak

detection equipment needs to be calibrated for the gas it is being used to detect. Therefore, there is a concern that leak detection devices might not appropriately detect or alarm for a blend; especially if there's variations in the concentrations of the blend.

Q. What, if any, limits should be placed on a pilot program prior to the introduction of hydrogen into an existing system?

A. The Panel recommends that any pilot program will need to adequately address these safety concerns and any other threats to the integrity of the pipeline caused by the introduction of hydrogen into the gas system.

Q. Does the Panel have any other concerns with respect to the HyGrid Project?

A. Yes, we are concerned with the expected in-service date for this project. According to the response to DPS-668, KEDLI states that it has only completed a 25 percent design report to date, has a new facility siting study in progress, and requires additional permits for the project. Given the stage of development of the project, the Companies' proposed in-service

date of September 2024 seems unlikely.

Q. What does the Panel recommend with respect to the in-service date of the HyGrid Project?

A. We recommend that the Commission delay the project for one year and exclude the requested FTEs from the revenue requirement for the Rate Year.

Q. If the project does go forward in the future, does the Panel have concerns with the proposed HyGrid Project?

A. Yes. We are concerned with project costs and the potential bill impact to customers and with how the Company will ensure that the pilot proceeds safely. The CAC Final Scoping Plan, on page 213, highlights the need to further study the use of hydrogen as an alternate fuel to help decarbonize difficult sectors. Further study of the use of hydrogen as a way to enable the continued viability of the gas infrastructure must be done in a way that does not compromise safety of the distribution system while limiting bill impacts to customers.

Q. What does the Panel recommend with regard to costs?

A. We recommend that, if it does allow the project to go forward, the Commission set a cap on the capital costs to those proposed in KEDLI___(GIOP-1). Any additional pilot costs above the cap should be funded by Shareholders. With regard to bill impacts to customers associated with the service area that is being included in the pilot, the Company should limit any commodity cost impacts to customers to the cost of natural gas and any risk of associated incremental costs should be borne by shareholders.

Q. What does the Panel recommend with regard to ensuring that the pilot proceeds safely?

A. We recommend that any hydrogen blending only occur in non-LPP, that is, plastic or protected steel segments. Also, the Companies should take affirmative steps to inspect and repair any and all known leaks prior to any blending on such segments.

Q. Does the Panel have any recommendations concerning the Companies' customer enrollment for the pilot?

A. No, we do not recommend any changes with the

Companies' proposed customer enrollment process.

Q. Does this conclude your testimony at this time?

A. Yes.